# 配置集群间 LIF

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# 配置集群间 LIF

## 在共享数据端口上配置集群间 LIF

您可以在与数据网络共享的端口上配置集群间 LIF 。这样可以减少集群间网络连接所需的端口数量。

步骤

1. 列出集群中的端口：

   ```
   network port show
   ```

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

   ```
   cluster01::> network port show
                                                                     Speed
   (Mbps)
   Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
   ------ --------- ------------ ---------------- ----- -------
   ------------
   cluster01-01
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
   cluster01-02
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
   ```

2. 在系统 SVM 上创建集群间 LIF ：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port port -address` *`port_ip`* `-netmask` *`netmask`* |

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |

有关完整的命令语法，请参见手册页。

以下示例将创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0c
-address 192.168.1.201
-netmask 255.255.255.0

cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0c
-address 192.168.1.202
-netmask 255.255.255.0
```

3. 验证是否已创建集群间 LIF：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
cluster01
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0c
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0c
true
```

4. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show – service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 `e0c` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 将故障转移到 `e0d` 端口。

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
cluster01
         cluster01_icl01 cluster01-01:e0c   local-only
192.168.1.201/24
                            Failover Targets: cluster01-01:e0c,
                                              cluster01-01:e0d
         cluster01_icl02 cluster01-02:e0c   local-only
192.168.1.201/24
                            Failover Targets: cluster01-02:e0c,
                                              cluster01-02:e0d
```

# 在专用端口上配置集群间 LIF

您可以在专用端口上配置集群间 LIF 。这样做通常会增加复制流量的可用带宽。

步骤

1. 列出集群中的端口：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

   ```
   cluster01::> network port show
                                                                  Speed
   (Mbps)
   Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
   ------ --------- ------------ ---------------- ----- -------
   ------------
   cluster01-01
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
          e0e       Default      Default          up     1500   auto/1000
          e0f       Default      Default          up     1500   auto/1000
   cluster01-02
          e0a       Cluster      Cluster          up     1500   auto/1000
          e0b       Cluster      Cluster          up     1500   auto/1000
          e0c       Default      Default          up     1500   auto/1000
          e0d       Default      Default          up     1500   auto/1000
          e0e       Default      Default          up     1500   auto/1000
          e0f       Default      Default          up     1500   auto/1000
   ```

2. 确定哪些端口可专用于集群间通信：

   `network interface show -fields home-port ， curr-port`

   有关完整的命令语法，请参见手册页。

   以下示例显示未为端口 `e0e` 和 `e0f` 分配 LIF ：

```
cluster01::> network interface show -fields home-port,curr-port
vserver lif                  home-port curr-port
------- -------------------- --------- ---------
Cluster cluster01-01_clus1   e0a       e0a
Cluster cluster01-01_clus2   e0b       e0b
Cluster cluster01-02_clus1   e0a       e0a
Cluster cluster01-02_clus2   e0b       e0b
cluster01
        cluster_mgmt         e0c       e0c
cluster01
        cluster01-01_mgmt1   e0c       e0c
cluster01
        cluster01-02_mgmt1   e0c       e0c
```

3. 为专用端口创建故障转移组：

   `network interface failover-groups create -vserver` *`system_svm`* `-failover-group` *`failover_group`* `-targets` *`physical_or_logical_ports`*

   以下示例将端口 `e0e` 和 `e0f` 分配给系统 SVM `cluster01` 上的故障转移组 `intercluster01`：

```
cluster01::> network interface failover-groups create -vserver cluster01
-failover-group
intercluster01 -targets
cluster01-01:e0e,cluster01-01:e0f,cluster01-02:e0e,cluster01-02:e0f
```

4. 验证是否已创建故障转移组：

   `network interface failover-groups show`

   有关完整的命令语法，请参见手册页。

```
cluster01::> network interface failover-groups show
                                  Failover
Vserver          Group            Targets
---------------- ----------------
--------------------------------------------
Cluster
                 Cluster
                                  cluster01-01:e0a, cluster01-01:e0b,
                                  cluster01-02:e0a, cluster01-02:e0b
cluster01
                 Default
                                  cluster01-01:e0c, cluster01-01:e0d,
                                  cluster01-02:e0c, cluster01-02:e0d,
                                  cluster01-01:e0e, cluster01-01:e0f
                                  cluster01-02:e0e, cluster01-02:e0f
                 intercluster01
                                  cluster01-01:e0e, cluster01-01:e0f
                                  cluster01-02:e0e, cluster01-02:e0f
```

5. 在系统 SVM 上创建集群间 LIF 并将其分配给故障转移组。

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* `-failover -group` *`failover_group`* |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* `-failover-group` *`failover_group`* |

有关完整的命令语法，请参见手册页。

以下示例将在故障转移组 `intercluster01` 中创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0e
-address 192.168.1.201
-netmask 255.255.255.0 -failover-group intercluster01

cluster01::> network interface create -vserver cluster01 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0e
-address 192.168.1.202
-netmask 255.255.255.0 -failover-group intercluster01
```

6. 验证是否已创建集群间 LIF：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node           Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
cluster01
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0e
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0f
true
```

7. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 SVM`e0e` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 将故障转移到 `e0f` 端口。

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
cluster01
         cluster01_icl01 cluster01-01:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-01:e0e,
                                               cluster01-01:e0f
         cluster01_icl02 cluster01-02:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-02:e0e,
                                               cluster01-02:e0f
```

# 在自定义 IP 空间中配置集群间 LIF

## 您可以在自定义 IP 空间中配置集群间 LIF 。这样可以在多租户环境中隔离复制流量。

创建自定义 IP 空间时，系统会创建一个系统 Storage Virtual Machine （ SVM ），用作该 IP 空间中系统对象的容器。您可以使用新的 SVM 作为新 IP 空间中任何集群间 LIF 的容器。新的 SVM 与自定义 IP 空间同名。

步骤

1. 列出集群中的端口：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示了 `cluster01` 中的网络端口：

```
cluster01::> network port show
                                                                  Speed
(Mbps)
Node   Port      IPspace      Broadcast Domain Link   MTU    Admin/Oper
------ --------- ------------ ---------------- ----- -------
------------
cluster01-01
       e0a       Cluster      Cluster          up     1500   auto/1000
       e0b       Cluster      Cluster          up     1500   auto/1000
       e0c       Default      Default          up     1500   auto/1000
       e0d       Default      Default          up     1500   auto/1000
       e0e       Default      Default          up     1500   auto/1000
       e0f       Default      Default          up     1500   auto/1000
cluster01-02
       e0a       Cluster      Cluster          up     1500   auto/1000
       e0b       Cluster      Cluster          up     1500   auto/1000
       e0c       Default      Default          up     1500   auto/1000
       e0d       Default      Default          up     1500   auto/1000
       e0e       Default      Default          up     1500   auto/1000
       e0f       Default      Default          up     1500   auto/1000
```

2. 在集群上创建自定义 IP 空间：

   `network ipspace create -ipspace` *`ipspace`*

   以下示例将创建自定义 IP 空间 `ipspace-IC1`：

   ```
   cluster01::> network ipspace create -ipspace ipspace-IC1
   ```

3. 确定哪些端口可专用于集群间通信：

   `network interface show -fields home-port ， curr-port`

   有关完整的命令语法，请参见手册页。

   以下示例显示未为端口 `e0e` 和 `e0f` 分配 LIF：

```
cluster01::> network interface show -fields home-port,curr-port
vserver lif                  home-port curr-port
------- -------------------- --------- ---------
Cluster cluster01_clus1   e0a       e0a
Cluster cluster01_clus2   e0b       e0b
Cluster cluster02_clus1   e0a       e0a
Cluster cluster02_clus2   e0b       e0b
cluster01
        cluster_mgmt         e0c       e0c
cluster01
        cluster01-01_mgmt1   e0c       e0c
cluster01
        cluster01-02_mgmt1   e0c       e0c
```

4. 从默认广播域中删除可用端口：

   `network port broadcast-domain remove-ports -broadcast-domain default -ports ports ports`

   一个端口一次不能位于多个广播域中。有关完整的命令语法，请参见手册页。

   以下示例将从默认广播域中删除端口 `e0e` 和 `e0f`：

```
cluster01::> network port broadcast-domain remove-ports -broadcast
-domain Default -ports
cluster01-01:e0e,cluster01-01:e0f,cluster01-02:e0e,cluster01-02:e0f
```

5. 验证端口是否已从默认广播域中删除：

   `network port show`

   有关完整的命令语法，请参见手册页。

   以下示例显示已从默认广播域中删除端口 `e0e` 和 `e0f`：

```
cluster01::> network port show
                                                          Speed (Mbps)
Node   Port    IPspace   Broadcast Domain Link   MTU    Admin/Oper
------ ------- --------- --------------- ----- ------- ------------
cluster01-01
       e0a     Cluster    Cluster          up    9000  auto/1000
       e0b     Cluster    Cluster          up    9000  auto/1000
       e0c     Default    Default          up    1500  auto/1000
       e0d     Default    Default          up    1500  auto/1000
       e0e     Default    -                up    1500  auto/1000
       e0f     Default    -                up    1500  auto/1000
       e0g     Default    Default          up    1500  auto/1000
cluster01-02
       e0a     Cluster    Cluster          up    9000  auto/1000
       e0b     Cluster    Cluster          up    9000  auto/1000
       e0c     Default    Default          up    1500  auto/1000
       e0d     Default    Default          up    1500  auto/1000
       e0e     Default    -                up    1500  auto/1000
       e0f     Default    -                up    1500  auto/1000
       e0g     Default    Default          up    1500  auto/1000
```

6. 在自定义 IP 空间中创建广播域：

   `network port broadcast-domain create -ipspace` *`ipspace`*`-broadcast-domain` *`broadcast_domain`* `-mtu` *`mtu _ -ports _ports`*

   以下示例将在 IP 空间 `ipspace-IC1` 中创建广播域 `ipspace-IC1-bd`：

```
cluster01::> network port broadcast-domain create -ipspace ipspace-IC1
-broadcast-domain
ipspace-IC1-bd -mtu 1500 -ports cluster01-01:e0e,cluster01-01:e0f,
cluster01-02:e0e,cluster01-02:e0f
```

7. 验证是否已创建广播域：

   `network port broadcast-domain show`

   有关完整的命令语法，请参见手册页。

```
cluster01::> network port broadcast-domain show
IPspace Broadcast                                         Update
Name    Domain Name    MTU  Port List                     Status Details
------- ----------- ------  ------------------------------ --------------
Cluster Cluster       9000
                            cluster01-01:e0a              complete
                            cluster01-01:e0b              complete
                            cluster01-02:e0a              complete
                            cluster01-02:e0b              complete
Default Default       1500
                            cluster01-01:e0c              complete
                            cluster01-01:e0d              complete
                            cluster01-01:e0f              complete
                            cluster01-01:e0g              complete
                            cluster01-02:e0c              complete
                            cluster01-02:e0d              complete
                            cluster01-02:e0f              complete
                            cluster01-02:e0g              complete
ipspace-IC1
        ipspace-IC1-bd
                      1500
                            cluster01-01:e0e              complete
                            cluster01-01:e0f              complete
                            cluster01-02:e0e              complete
                            cluster01-02:e0f              complete
```

8. 在系统 SVM 上创建集群间 LIF 并将其分配给广播域：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-service -policy default-intercluster -home -node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface create -vserver` *`system_svm`* `-lif` *`LIF_name`* `-role intercluster -home-node` *`node`*`-home-port` *`port`* `-address` *`port_ip`* `-netmask` *`netmask`* |

此时将在分配主端口的广播域中创建 LIF 。此广播域具有一个默认故障转移组，其名称与此广播域相同。有关完整的命令语法，请参见手册页。

以下示例将在广播域 `ipspace-IC1-bd` 中创建集群间 LIF `cluster01_icl01` 和 `cluster01_icl02`：

```
cluster01::> network interface create -vserver ipspace-IC1 -lif
cluster01_icl01 -service-
policy default-intercluster -home-node cluster01-01 -home-port e0e
-address 192.168.1.201
-netmask 255.255.255.0

cluster01::> network interface create -vserver ipspace-IC1 -lif
cluster01_icl02 -service-
policy default-intercluster -home-node cluster01-02 -home-port e0e
-address 192.168.1.202
-netmask 255.255.255.0
```

9. 验证是否已创建集群间 LIF：

| 选项 | Description |
|---|---|
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster` |

有关完整的命令语法，请参见手册页。

```
cluster01::> network interface show -service-policy default-intercluster
            Logical    Status     Network            Current
Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node           Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
ipspace-IC1
            cluster01_icl01
                       up/up      192.168.1.201/24   cluster01-01  e0e
true
            cluster01_icl02
                       up/up      192.168.1.202/24   cluster01-02  e0f
true
```

10. 验证集群间 LIF 是否冗余：

| 选项 | Description |
| --- | --- |
| • 在 ONTAP 9.6 及更高版本中： * | `network interface show -service-policy default-intercluster -failover` |
| • 在 ONTAP 9.5 及更早版本中： * | `network interface show -role intercluster -failover` |

有关完整的命令语法，请参见手册页。

以下示例显示 SVM `e0e` 端口上的集群间 LIF `cluster01_icl01` 和 `cluster01_icl02` 故障转移到`e0f`端口：

```
cluster01::> network interface show -service-policy default-intercluster
-failover
         Logical         Home                  Failover        Failover
Vserver  Interface       Node:Port             Policy          Group
-------- --------------- --------------------- --------------- --------
ipspace-IC1
         cluster01_icl01 cluster01-01:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-01:e0e,
                                               cluster01-01:e0f
         cluster01_icl02 cluster01-02:e0e   local-only
intercluster01
                            Failover Targets:  cluster01-02:e0e,
                                               cluster01-02:e0f
```

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